

Table 3: **Trustworthy Transfer Learning Results.** Ent, MaxP, MI, and Dent stand for different uncertainty measurements, i.e., Entropy, Max Probability, Mutual Information, and Differential Entropy, respectively. We use ResNet-50 pretrained with ImageNet as the source model and FashionMNIST as OOD samples.

Methods	Target	Test Acc	AUROC	AUPR	Target	Test Acc	AUROC	AUPR
FineTune(Ent)	STL10	48.1±0.5	89.2±0.9	89.3±1.2	CIFAR10	65.0±0.4	74.8±1.6	71.6±1.8
FineTune(MaxP)		48.1±0.5	81.8±1.2	83.2±1.7		65.0±0.4	72.7±1.4	69.4±1.5
CrossEnt Loss(Ent)		48.0±0.2	88.9±1.0	87.4±1.5		86.3±0.1	85.0±1.0	82.2±1.9
CrossEnt Loss(MaxP)		48.0±0.2	84.9±0.7	84.7±0.8		86.3±0.1	83.1±0.9	79.0±1.6
Ours(Ent)		47.2±0.3	91.8±0.8	91.3±0.7		86.6±0.3	89.9±1.3	88.8±1.5
Ours(MaxP)		47.2±0.3	87.3±1.7	87.9±1.5		86.6±0.3	87.6±1.6	85.5±2.0
Ours(MI)		47.2±0.3	90.2±2.0	88.7±2.8		86.6±0.3	90.8±0.9	89.9±1.2
Ours(Dent)		47.2±0.3	91.7±1.0	90.3±1.7		86.6±0.3	92.0±0.8	91.1±0.8

as “**LastLayer**”. It turns out even such a naive method can achieve improved performance compared to the base model, which further justifies the effectiveness of the post-hoc uncertainty learning setting, i.e., the benefit of solely focusing on UQ performance at the second stage. This interesting observation inspires us to conjecture that efficiently retraining the classifier of the base model at the second stage will lead to better UQ performance. A theoretical investigation of this observation can be interesting for future work.

Data Efficiency. Instead of using all the training samples, we randomly choose only 10% samples to train the meta-model. The results are shown in Table 4 as “10%**data**”. It can be observed that our meta-model requires a small amount of data to achieve comparable performance due to the smaller model complexity. Therefore, our proposed method is also more computationally efficient than the approaches that retrain the whole model from scratch.

Table 4: **Ablation Study of Meta-model (CIFAR10 AUROC score).** The results are reported as mean over five experiment trails. Error bars are provided in Table ?? in the Appendix.

Methods	SVHN	FMNIST	LSUN	TIM	Corrupted
Base Model(Ent)	86.4	90.8	89.0	87.5	85.9
Base Model(MaxP)	86.3	90.4	88.7	87.3	85.7
Linear-Meta(Ent)	90.4	91.3	91.5	89.5	90.4
Linear-Meta(MaxP)	90.1	91.5	91.4	89.6	90.1
Linear-Meta(MI)	90.6	90.1	91.2	88.8	90.5
Linear-Meta(Dent)	90.4	90.7	91.4	89.2	90.3
Linear-Meta(Prec)	90.6	90.0	91.2	88.8	90.6
Cross-Ent(Ent)	94.2	91.2	91.2	90.3	94.7
Cross-Ent(MaxP)	93.3	91.1	90.9	90.0	94.0
LastLayer(Ent)	93.0	90.2	91.9	89.9	93.1
LastLayer(MaxP)	92.9	90.5	91.9	90.1	93.1
10% data(Ent)	90.0	89.1	88.7	88.2	90.2
10% data(MaxP)	90.9	88.1	86.9	86.5	91.7
10% data(MI)	99.9	98.1	95.4	97.2	99.9
10% data(Dent)	96.7	97.4	94.3	95.7	96.7
10% data(Prec)	99.9	98.0	95.4	97.3	99.9
Ours(Ent)	96.3	89.0	89.6	89.4	95.9
Ours(MaxP)	95.6	87.8	89.1	88.2	94.0
Ours(MI)	100.0	98.8	95.2	98.1	100.0
Ours(Dent)	100.0	98.4	95.7	97.7	100.0
Ours(Prec)	100.0	98.8	95.1	98.1	100.0

6 Concluding Remarks

We provide a novel solution for the uncertainty quantification problem via our proposed post-hoc uncertainty learning framework and the Dirichlet meta-model approach. Our method turns out to be both effective and computationally efficient for various UQ applications. We believe our meta-model approach not only has the flexibility to tackle other applications relevant to uncertainty quantification, such as quantifying transfer-ability in transfer learning and domain adaptation, but also can adapt to other model architecture such as transformer and language model. Exploring these potential applications and offering a theoretical interpretation of the meta-model can be interesting future work.

Acknowledgement

This work was supported, in part, by the MIT-IBM Watson AI Lab under Agreement No. W1771646, and NSF under Grant No. CCF-1816209.

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